2014-DSE
BIO
PAPER 1B

B

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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2014

Candidate Number									

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) Present your answers in paragraphs wherever appropriate.

(7) The diagrams in this section are **NOT** necessarily drawn to scale.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

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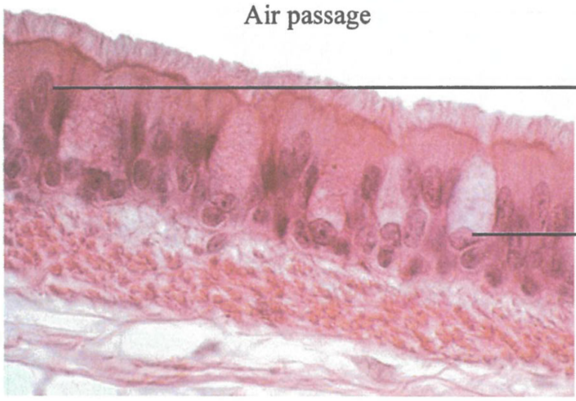
SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. For each of the biomolecules listed in Column 1, select from Column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1		Column 2	
NADPH	______	A.	a product of oxidative phosphorylation
pyruvate	______	B.	a product of photochemical reactions
NAD	______	C.	a product of carbon fixation
		D.	a product of glycolysis

2. The photomicrograph below shows a section of the inner wall of the human trachea:



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(x 400)

(a) In the spaces provided, label the cells shown in the photomicrograph. (2 marks)

(b) With reference to the features of the inner wall shown in the photomicrograph, describe how the inner wall of the trachea can protect our body against bacterial invasion. (3 marks)

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Answers written in the margins will not be marked.

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3. Before the early 20th century, scientists generally held the belief:

"Cell division resulted in the loss of genetic material so that each cell in a multicellular organism would contain only the genetic material specific to its particular cell type."

In 1902, Hans Spemann performed one of the earliest experiments on animal cloning. He used a fine hair to separate the cells of a two-celled amphibian embryo, and found that each cell was able to develop into a complete organism.

(a) Why did Spemann's experiment disprove the early belief about cell division? (1 mark)

(b) Elaborate on how the above example can be used to demonstrate the two aspects of the nature of science listed in the table below. (2 marks)

Nature of Science	*Elaboration*
Scientific knowledge is tentative and subject to change.	
Interpretation of observations is guided by our prior understanding of other theories and concepts.	

(c) Using the current understanding about cell division, explain how genetic material is preserved in mitosis. (3 marks)

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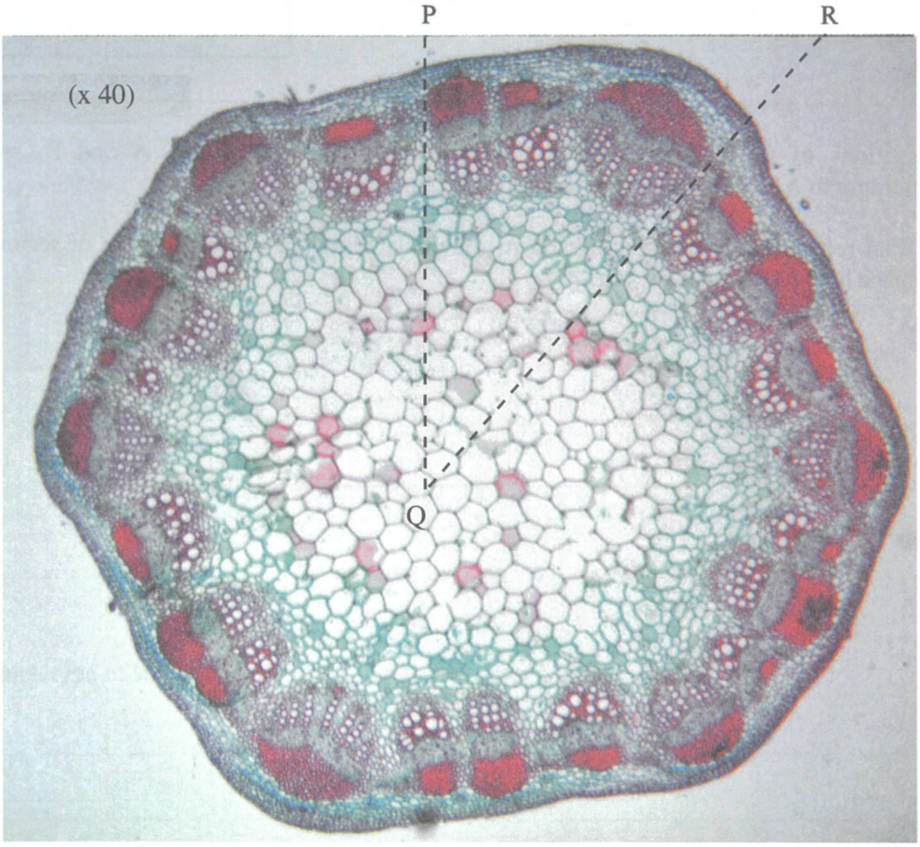
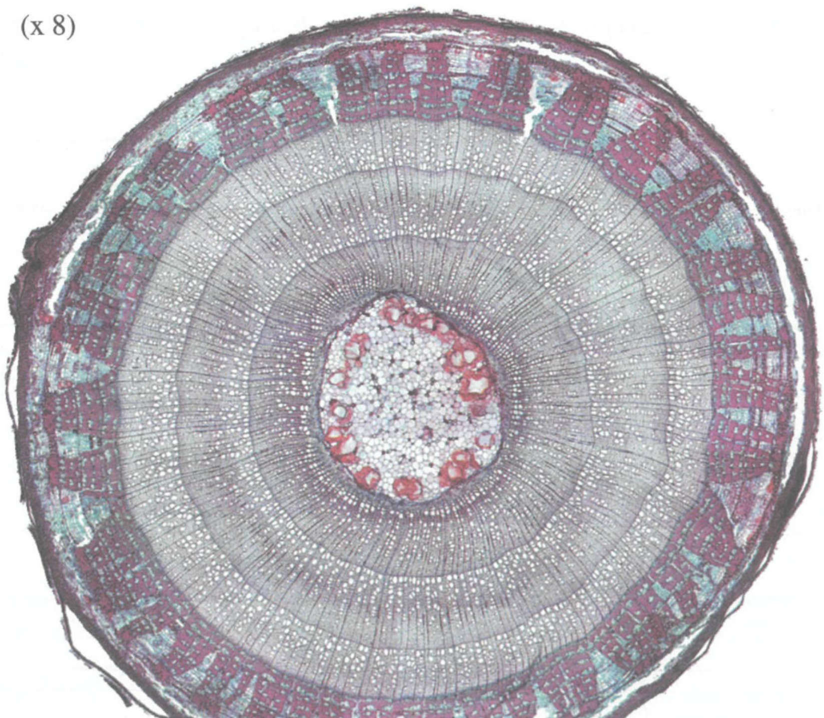
4. ***Photomicrograph A – Stem of plant A***



Photomicrograph B – Stem of plant B



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Answers written in the margins will not be marked.

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4. Cross sections of the stems from two different dicotyledonous plants, A and B, are shown in Photomicrograph A and Photomicrograph B on the opposite page.

(a) With reference to Photomicrograph A, draw a labelled low-power diagram of sector PQR in the space below. (5 marks)

Answers written in the margins will not be marked.

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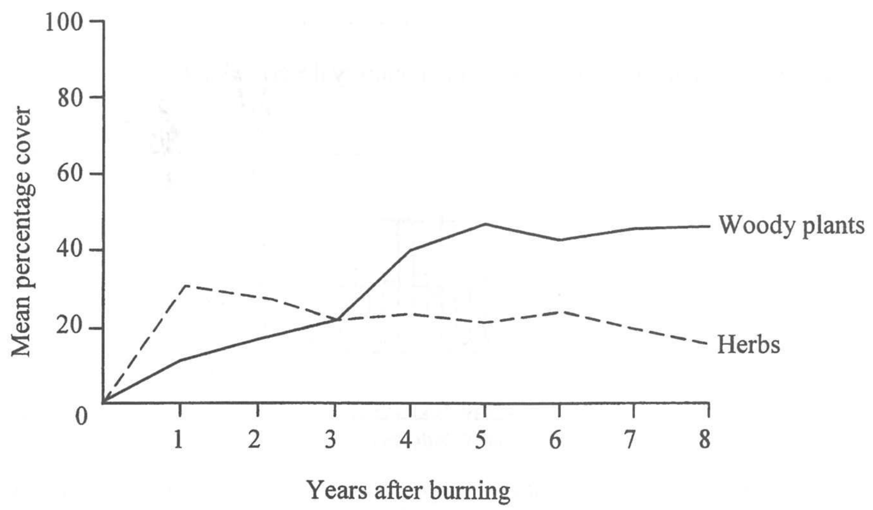
(b) With reference to the photomicrographs, deduce the major means of support in plants A and B. (4 marks)

Answers written in the margins will not be marked.

5. In a study, a plot of land was burnt by fire. After this, the vegetation on this land, classified into herbs and woody plants, was monitored for 8 years. The percentage cover of each type of vegetation is shown in the graph below:



(a) Which type of succession is shown in the above case? Explain your answer. (2 marks)

(b) (i) Describe briefly how the dominant community of vegetation changes with time after the fire. (2 marks)

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(ii) Explain the changes in the dominant community described in (i). (4 marks)

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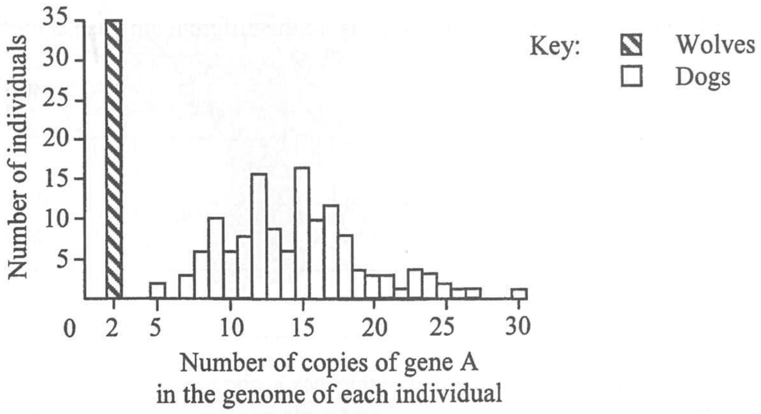
6. It is generally believed that domestic dogs evolved from ancient wolves. A recent study comparing the genomes of wolves and dogs suggests that genes with key roles in starch digestion were selected during the domestication of wolves into dogs. One of these genes was gene A, which codes for amylase. This gene may exist in many copies in a genome. The following graph shows the number of individuals having different numbers of copies of gene A in 35 wolves and 136 dogs:



(a) Based on the data above and the gene expression processes, explain why the amylase activity in dogs is generally higher than that in wolves. (3 marks)

(b) It is hypothesized that in ancient times, wolves might have been attracted to waste dumps near early human settlements and consumed human food waste. Suggest how the domestication of wolves would have led to the selection of multiple copies of gene A. (5 marks)

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(c) Describe an experiment which can compare the different amylase activities of wolves and dogs. (4 marks)

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